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Basic Terminologies

Lecture 3

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Introduction to Cryptography

Key Concepts:

- **Plaintext:** Original message before encryption.
  - Example: "HELLO"
- **Ciphertext:** Coded message after encryption.
  - Example: "URYYB" (using Caesar Cipher with key = 13)
- **Cipher:** Algorithm for transforming plaintext into ciphertext.
  - Example: Caesar Cipher shifts each letter by a certain number.

Key and Encryption/Decryption

- **Key:** Information used in the cipher, known only to the sender and receiver.

Example: In Caesar Cipher, the key is the number of positions to shift each letter (e.g., key = 3).

- **Encipher (Encrypt):** Converting plaintext to ciphertext.

Example: "HELLO" becomes "KHOOR" using a Caesar Cipher with key = 3.

- **Decipher (Decrypt):** Recovering plaintext from ciphertext.

Example: "KHOOR" decrypted back to "HELLO" with the same key = 3.

Cryptography vs Cryptanalysis

- **Cryptography:** Study of encryption methods.

Example: Symmetric-key encryption (AES) and public-key encryption (RSA).

- **Cryptanalysis (Codebreaking):** Study of methods to decipher ciphertext without knowing the key.

Example: Frequency analysis used to break simple substitution ciphers.

- **Cryptology:** Field encompassing both cryptography and cryptanalysis.

What is Social Engineering?

Social Engineering is the act of manipulating people into revealing confidential information or performing actions that could lead to security breaches. It takes advantage of human psychology rather than technical vulnerabilities.

OR

Social Engineering is a method used by attackers to trick individuals into giving away confidential information, usually through deception and manipulation.

What is Social Engineering?

Example:

Scenario: A hacker calls an employee, pretending to be from the company's IT department. They claim there's an urgent issue and ask the employee to reset their password. The employee follows the instructions and shares their new password over the phone.

Result: The hacker now has access to the employee's account, potentially leading to a security breach.

Types of Social Engineering Attacks:

- **Phishing**: Fake emails or websites that trick people into revealing passwords.
- **Pretexting**: Creating a fabricated scenario to get information.
- **Baiting**: Offering something tempting (e.g., free USB drives) to trick people into exposing their system.

How to Prevent It:

- Be cautious of unsolicited requests for information.
- Verify the identity of anyone asking for sensitive information.
- Don’t share passwords or personal information without confirmation.

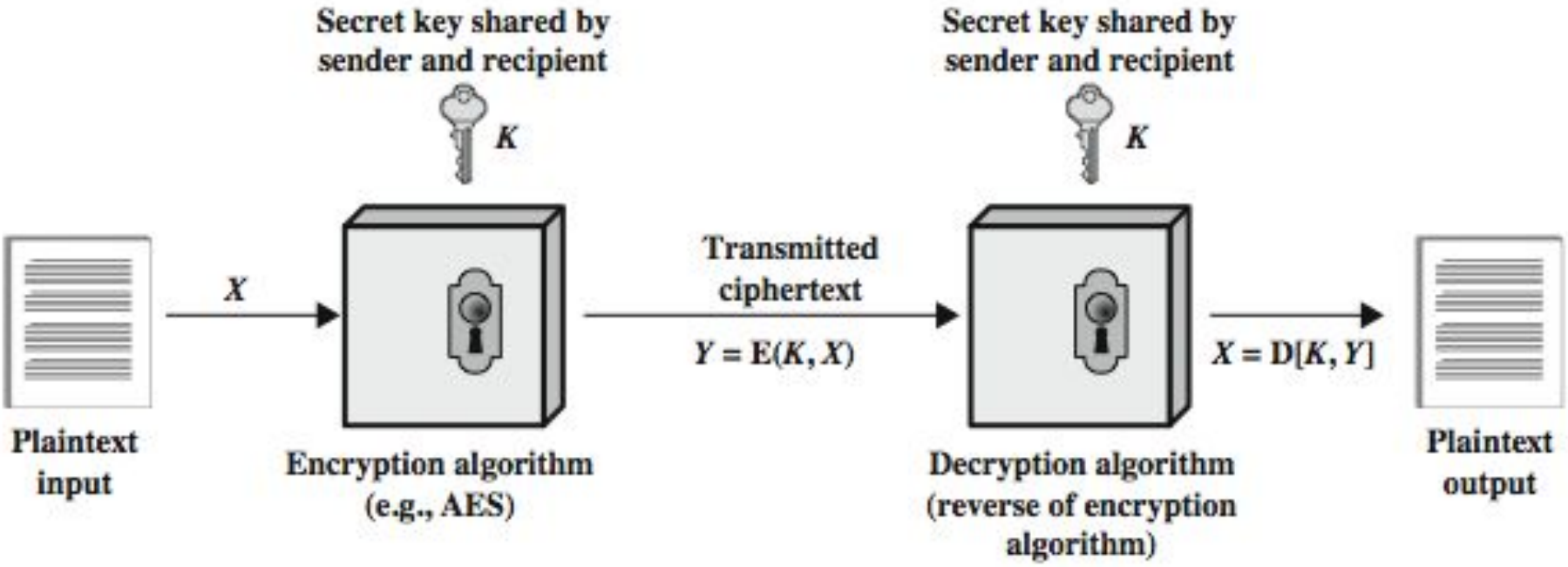
CODE & DECODE

- An original message is known as the plaintext, while the coded message is called the cipher text. The process of converting from plaintext to cipher text is known as enciphering or encryption.
- Restoring the plaintext from the cipher text is deciphering or decryption.
- The many schemes used for encryption constitute the area of study known as cryptography.
- Techniques used for deciphering a message without any knowledge of the enciphering details fall into the area of cryptanalysis.
- Cryptanalysis is what the layperson calls “breaking the code.”
- The areas of cryptography and cryptanalysis together are called cryptology.



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Symmetric Cipher Model



Symmetric Cipher Model

A symmetric encryption scheme has five ingredients

Plaintext: This is the original, understandable message or data that you start with before any processing.

Encryption Algorithm: This algorithm takes the plaintext and performs a series of changes and substitutions to make it unreadable.

Secret Key: This is an additional input that goes into the encryption process. It's a separate value from the plaintext and the algorithm itself. The algorithm uses this key to determine how it will change the plaintext. Using different keys will result in different scrambled outputs even if the original message is the same.

Symmetric Cipher Model

Ciphertext: This is the jumbled, unreadable version of the original message that comes out after encryption. It looks like random data and doesn't make sense on its own. The specific key used during encryption will produce a unique ciphertext for the same message.

Decryption Algorithm: This process is essentially the encryption algorithm in reverse. It takes the scrambled ciphertext and the secret key, and turns it back into the original readable message (plaintext).

ATTACKS

- The main objective of attacking an encryption system is to recover the key.
- Two Approaches for attacking a conventional encryption scheme:

CRYPTANALYSIS:

- Rely on the nature of the algorithm
- knowledge of the general characteristics of the plaintext
- plaintext–ciphertext pairs

BRUTE-FORCE ATTACK:

- Attacker tries every possible key on a piece of ciphertext until plaintext is obtained.
- Half of all possible keys must be tried to achieve success.

Types of Cryptanalysis

- **Ciphertext-only Attack:** The attacker only has access to the ciphertext.

  Example: An attacker intercepts an encrypted message and tries to break it using patterns or frequency analysis.
- **Known-plaintext Attack**: The attacker has both the plaintext and its corresponding ciphertext.

  Example: If an attacker knows the phrase "HELLO" corresponds to "KHOOR" (in a Caesar Cipher), they can work out the key (shift of 3).

Types of Cryptanalysis

- **Chosen-plaintext Attack:** The attacker can encrypt plaintext of their choice and study the resulting ciphertext.

  Example: Sending specific messages to an encryption system and analyzing how they get transformed.

- **Differential and Linear Cryptanalysis:** Advanced techniques used to break modern cryptosystems by analyzing how small changes in input affect the output.

Brute Force Attack

A brute force attack is a cryptanalytic method where the attacker systematically tries all possible keys or passwords until the correct one is found.

How It Works:

- The attacker generates every possible combination of letters, numbers, and symbols until the right key is found.
  - Example: In a Caesar Cipher, an attacker could try all possible shifts (from 1 to 25) to decrypt the ciphertext.

Brute Force Attack

Example:

- **Password Cracking:**
  - If the password is a simple combination of 4 digits, the brute force attacker tries all combinations from "0000" to "9999".
  - If the correct password is "4679", the attacker eventually finds it after checking 4679 combinations.

Brute Force Attack

Why Brute Force Attacks Work:

- They are effective against weak ciphers or short passwords.
- The longer and more complex the password, the harder and more time-consuming the attack.

Types of Brute Force Attack

Simple Brute Force Attacks: Trying all possible combinations of passwords until the correct one is found.

Dictionary Attacks: Using a pre-defined list of words (like a dictionary) to guess the password.

Hybrid Brute Force Attacks: Combining dictionary attacks with random variations (e.g., adding numbers or symbols).

Reverse Brute Force Attacks: Using a known password to find matching usernames or accounts.

Credential Stuffing: Using stolen usernames and passwords from one service to try and access other services.

Brute Force Attack

Countermeasures:

1. **Use Strong Passwords**: Include a mix of uppercase, lowercase, symbols, and numbers.
2. **Increase Key Length**: Larger key sizes in encryption (e.g., AES-256) make brute-force attacks practically impossible due to time constraints.
3. **Implement Lockout Mechanisms**: After several incorrect attempts, lock the system to prevent further attacks.

Questions